

Malignant Skin Lesion Detection Model using YOLOv8

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1. Objective

The objective of this project is to develop an AI-based object detection model that can identify and localize malignant skin lesion regions from medical images using YOLOv8.

The project demonstrates the complete computer vision pipeline: dataset preparation, model training, evaluation, and prediction.

2. Problem Statement

Manual examination of skin lesions requires expert analysis and can be time-consuming. An automated detection system can assist healthcare workflows by providing AI-based screening support and highlighting suspicious lesion areas.

3. Dataset Description

Dataset: Roboflow Skin Lesion Dataset

Task: Object Detection

Format: YOLOv8 Object Detection format

Classes:

- Malignant skin lesion

Dataset Split:

- Training images: 613
- Validation images: 99
- Testing images: 47

4. Dataset Visualization

A sample dataset image with annotations/bounding boxes.



5. Model and Approach

Model:

YOLOv8n Object Detection

YOLOv8n is a lightweight object detection model pretrained on the COCO dataset. The model was fine-tuned on the skin lesion dataset to detect malignant lesion regions using bounding box predictions.

6. Training Configuration

Model: YOLOv8n

Task: Object Detection

Epochs: 50

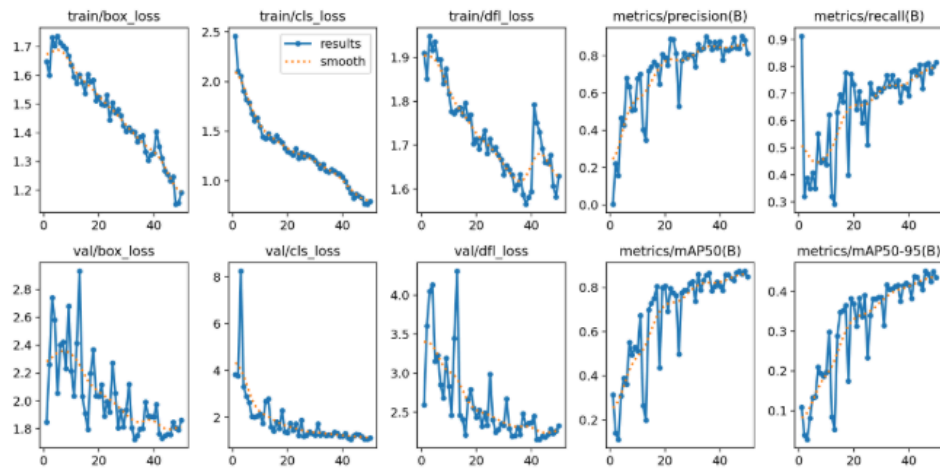
Image Size: 640

Batch Size: 16

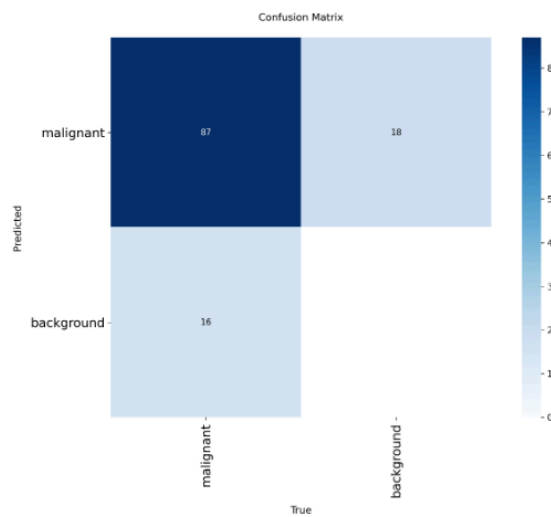
Framework: Ultralytics YOLO

Environment: Google Colab GPU

8. Model Evaluation



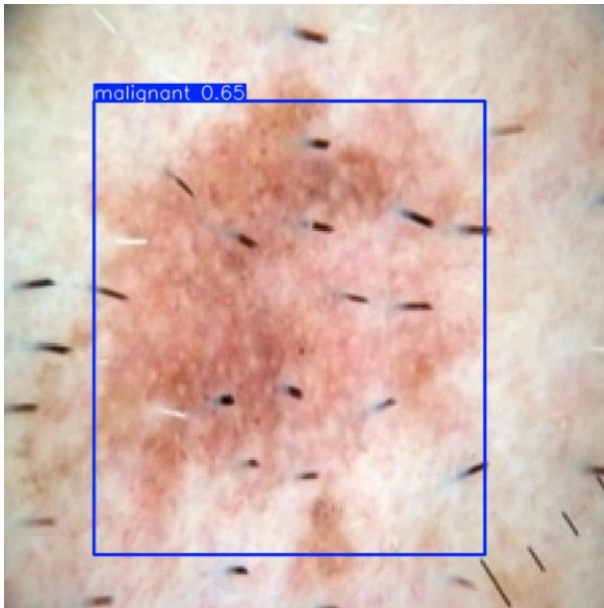
9. Confusion Matrix



10. Model Testing and Predictions

The trained model was tested on unseen test images.

The prediction images show detected lesion regions along with confidence scores.



11. Real World Application

Possible applications:

- AI-assisted dermatology screening
- Faster preliminary skin lesion analysis
- Supporting healthcare professionals
- Automated medical image analysis

12. Conclusion

The YOLOv8 object detection model was successfully trained on the skin lesion dataset. The model learned visual patterns and was able to localize malignant skin lesion regions using bounding box predictions.

The project demonstrated the complete computer vision workflow:
dataset preparation → YOLOv8 training → evaluation → prediction.

Files included:

- Training notebook
- Best trained model weights (best.pt)
- Training graphs
- Confusion matrix
- Prediction samples